CHAPTER 2

Basic Structures: Sets, Functions, Sequences, and Sums

2.1 Sets

2.2 Set Operations

2.3 Functions

2.4 Sequence and Summations

2.5 Cardinality of Sets



【Definition】 Let A and B be nonempty sets. A *function* (*mapping* or *transformations*) f from A to B:

$$f : A \to B$$

$$\forall a(a \in A \to \exists! b\,(b \in B \wedge f(a)=b))$$

A is called the *domain* of f, B is called the *codomain* of f.

$f(a) = b$

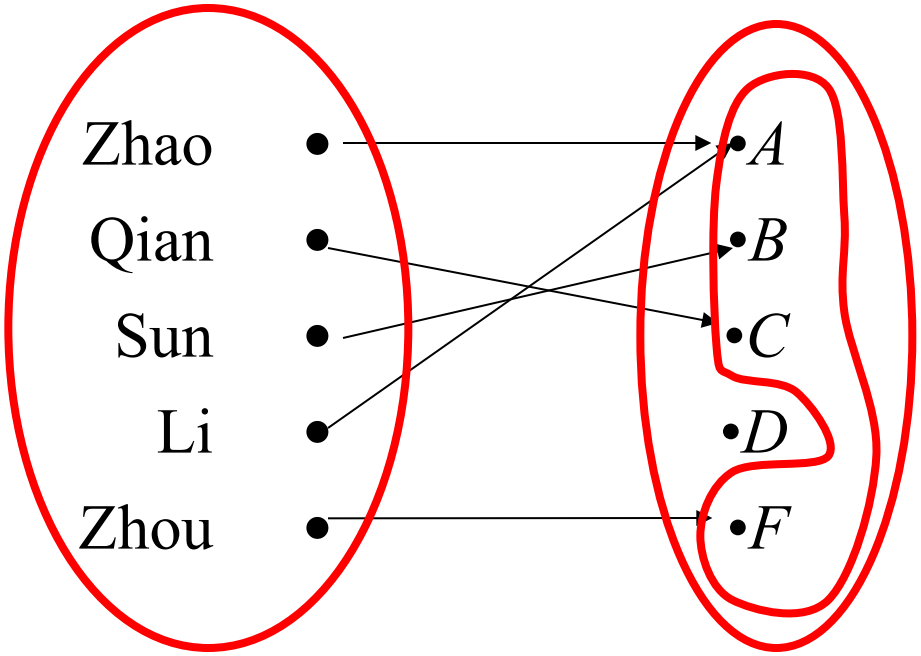
- b is called the *image* of a under f
- a is called a *preimage* of b

$f(A)$: the *range* of f is the set of all images of elements in A under f.

If f is a function from A to B, we say that f *maps* A to B.



〖**Example 1**〗 Suppose that each student in a class is assigned a letter grade from the set $\{A,B,C,D,F\}$. Let g be the function that assigns a grade to a student.



Domain

Codomain

Range

【Definition】 Let f_1 and f_2 be functions from A to R. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to R defined by

$$(f_1 + f_2)(x) = f_1(x) + f_2(x)$$

$$(f_1 f_2)(x) = f_1(x)\, f_2(x)$$

【Definition】 Let f be a function from A to B and let S be a subset of A. The image of S is the subset of B that consists of the images of the elements of S. We denote the image of S by $f(S)$, so that

$$f(S) = \{ f(s) \mid s \in S \}$$

- $f(\varnothing) = \varnothing$
- $f(\{a\}) = \{f(a)\}$
- $f(A \cup B) = f(A) \cup f(B)$
- $f(A \cap B) \subseteq f(A) \cap f(B)$

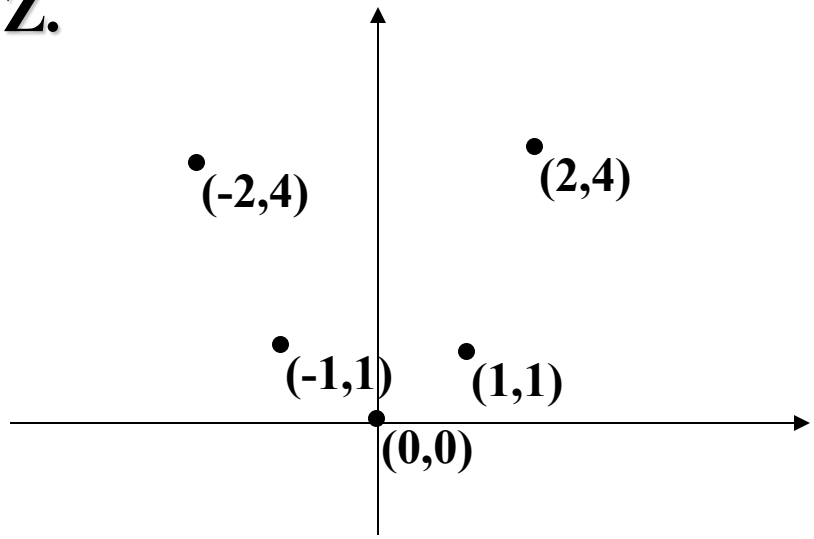


The Graphs of Functions

Let f be a function from the set A to the set B. The *graph* of the function f is the set of ordered pairs

$$\{(a, b) \mid a \in A \text{ and } f(a) = b\}.$$

〖Example 2〗 Display the graph of the function $f(x) = x^2$ from Z to Z.



One-to-One and Onto Functions

1) One-to-One Functions

A function *f* is *one-to-one* (denoted 1-1), or *injective*

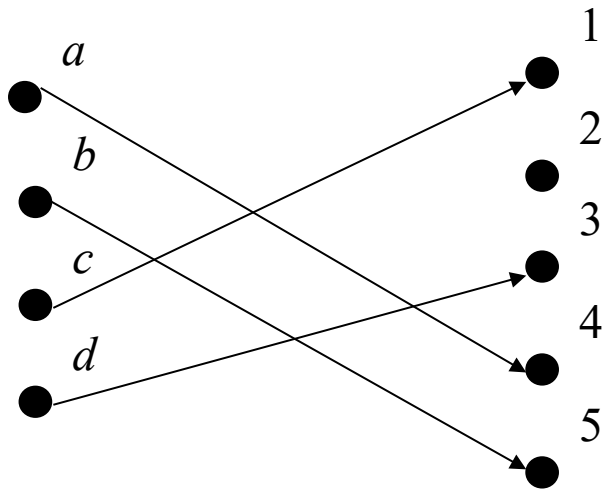
$$\forall a \forall b(f(a) = f(b) \rightarrow a = b)$$

Note:

- **This means that if $x \neq y$ then $f(x) \neq f(y)$.**
- **A function is said to be an *injection* if it is 1-1.**



〖Example 3〗Determine whether the function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4, 5\}$ with $f(a) = 4$, $f(b) = 5$, $f(c) = 1$, and $f(d) = 3$ is one-to-one.



Questions:

$f(x) = x^2$ from the set of integers to the set of integers X

$f(x) = x+1$ from the set of integers to the set of integers √

2) Onto Functions

A function *f* from *A* to *B* is called *onto*, or *surjective*

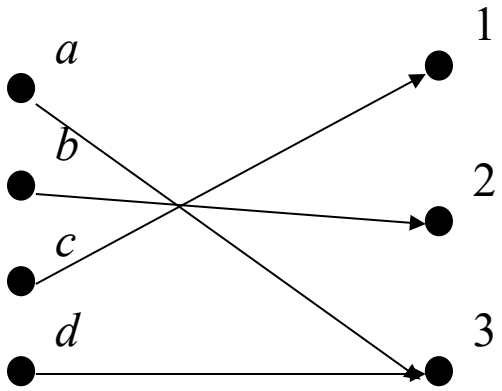
$$\forall b \in B \exists a \in A(f(a) = b)$$

Note:

- **This means that for every *b* in *B* there must be an *a* in *A* such that *f*(*a*) = *b*.**
- **Every *b* in *B* has a preimage.**
- **A function is called a *surjection* if it is onto.**



〖Example 4〗 Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3, f(b) = 2, f(c) = 1$, and $f(d) = 3$. Is f an onto function?



Questions:

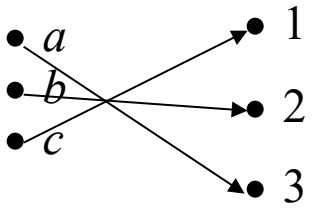
$f(x) = x^2$ from the set of integers to the set of integers X

$f(x) = x+1$ from the set of integers to the set of integers √

3) One-to-one Correspondence Functions

The function *f* is a *one-to-one correspondence*, or a *bijection*, if it is both *one-to-one* and *onto*.

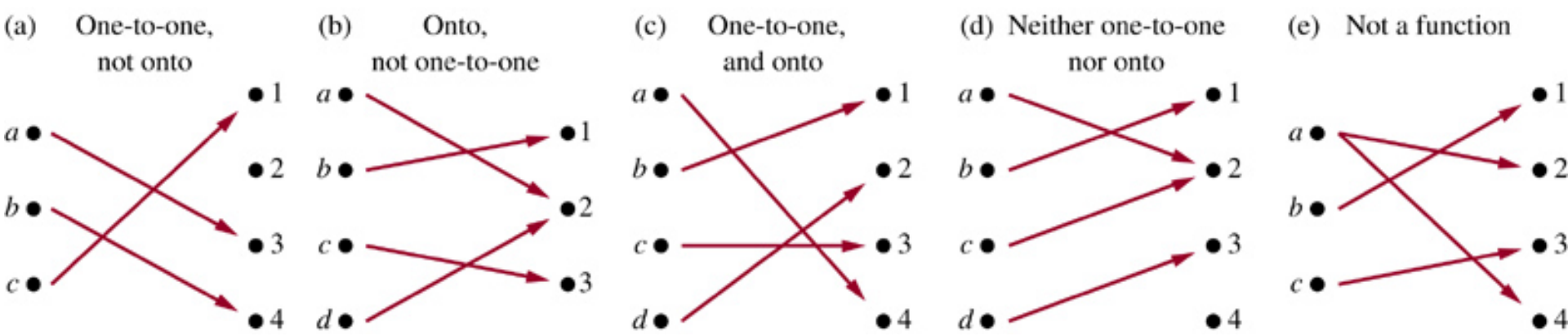
For example,



Note:

- **Whenever there is a bijection from *A* to *B*, the two sets must have the same number of elements or the same *cardinality*.**

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Showing that *f* is one-to-one or onto

Suppose that $f : A \to B$.

To show that f is injective Show that if $f(x) = f(y)$ for arbitrary $x, y \in A$, then $x = y$.

To show that f is not injective Find particular elements $x, y \in A$ such that $x \neq y$ and $f(x) = f(y)$.

To show that f is surjective Consider an arbitrary element $y \in B$ and find an element $x \in A$ such that $f(x) = y$.

To show that f is not surjective Find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.



Monotonic Functions

A monotonic function *f* is

- **either *monotonically* (*strictly*) *increasing***

$$\forall x \forall y (x < y \rightarrow f(x) < f(y))$$

- **or *monotonically* (*strictly*) *decreasing***

$$\forall x \forall y (x < y \rightarrow f(x) > f(y))$$



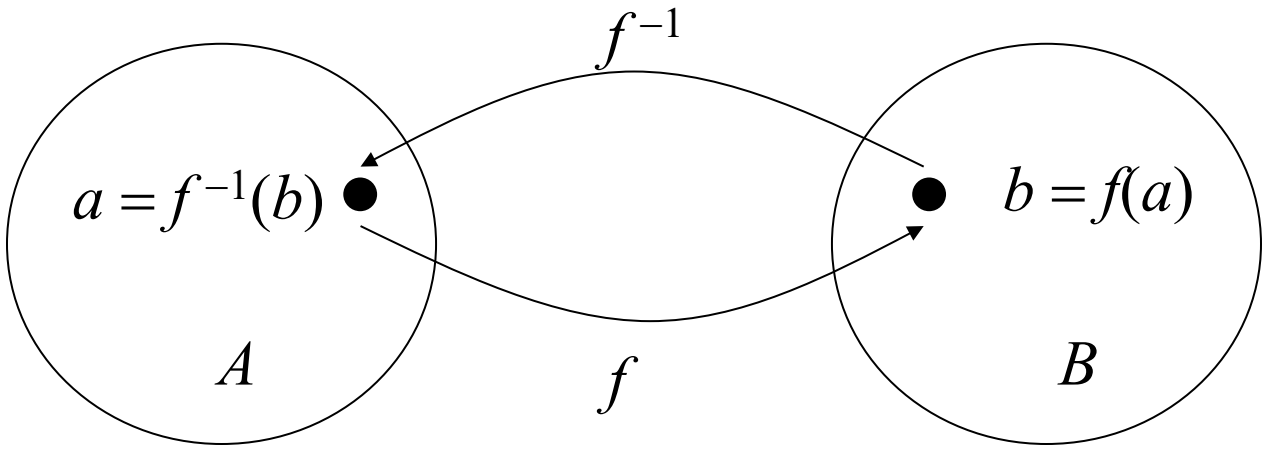
Inverse Functions

Let f be a bijection from A to B. Then the *inverse function* of f, denoted as f^{-1}, is the function from B to A defined as

$$f^{-1}(y) = x \text{ iff } f(x) = y$$

Note:

- **No inverse function exists unless f is a bijection.**
- **Function f is invertible if f f is bijective**



〖Example 5〗Let f be the function from the set of integers to the set of integers such that $f(x) = x+1$. what is its inverse?

f is one-to-one √

onto √

bijection √

$f^{-1}(y) = y - 1$

〖Example 6〗Let f be the function from Z to Z with $f(x) = x^2$. Is f invertible?

f is one-to-one X

No inverse function



Compositions of Functions

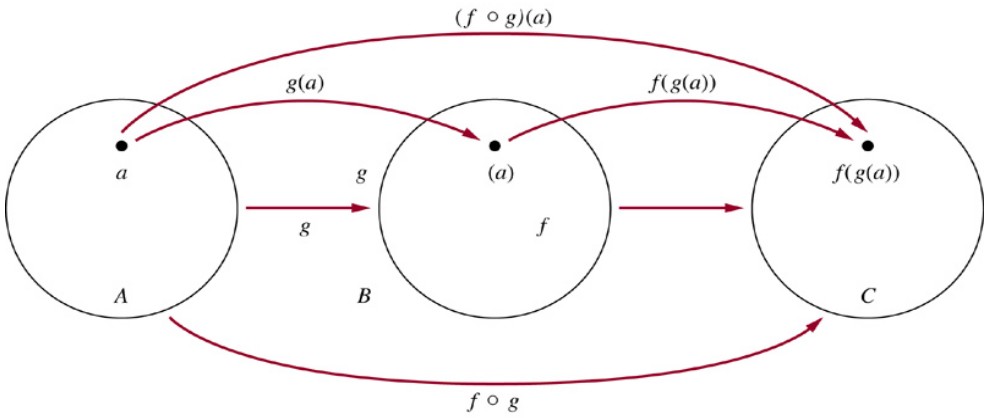
Let g be a function from the set A to the set B and let f be a function from the set B to the set C. The *composition* of the functions f and g, denoted by $f \circ g$, is defined by:

$$f \circ g(a) = f(g(a))$$

Note:

- **$f \circ g$ can't be defined unless the range of g is a subset of the domain of f.**

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〖Example 5〗 $f(x) = 2x + 3,\ g(x) = 3x + 2.$

What is the composition of f and g?

What is the composition of g and f?

$$f \circ g(x) = f(g(x)) = f(3x+2)$$
$$= 2(3x+2)+3 = 6x+7$$
$$g \circ f(x) = g(f(x)) = g(2x+3)$$
$$= 3(2x+3)+2 = 6x+11$$



5. Some Important Functions

- **The floor function**
- **The ceiling function**



【Definition】The *floor function* $f(x)$ is the largest integer less than or equal to the real number x.

Notation:

$\lfloor x \rfloor$

The floor function is often called the *greatest integer function*. It is often denoted by $[x]$.

For example:

$\lfloor 0.5 \rfloor = 0$

$\lfloor 1.5 \rfloor = 1$

$\lfloor 2 \rfloor = 2$

$\lfloor -0.5 \rfloor = -1$



【Definition】The *ceiling function* $f(x)$ is the smallest integer greater than or equal to the real number x.

Notation:

$$\lceil x \rceil$$

For example:

$$\lceil 0.5 \rceil = 1$$

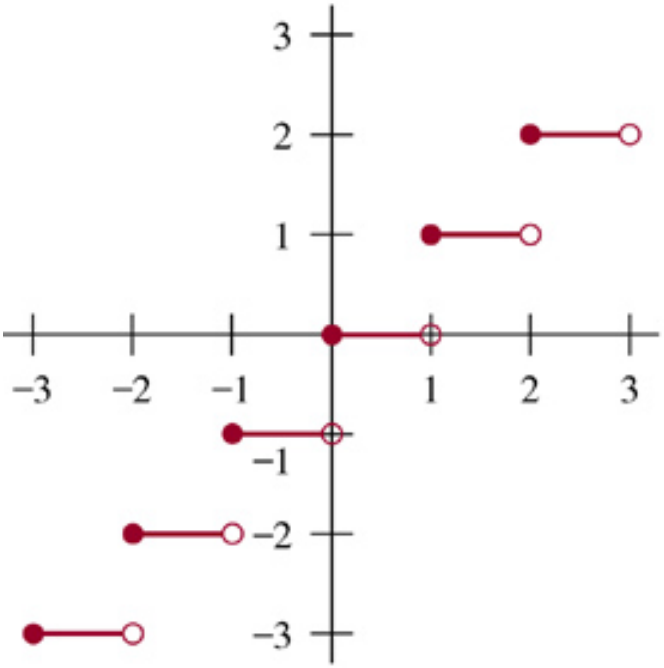
$$\lceil 1.5 \rceil = 2$$

$$\lceil 2 \rceil = 2$$

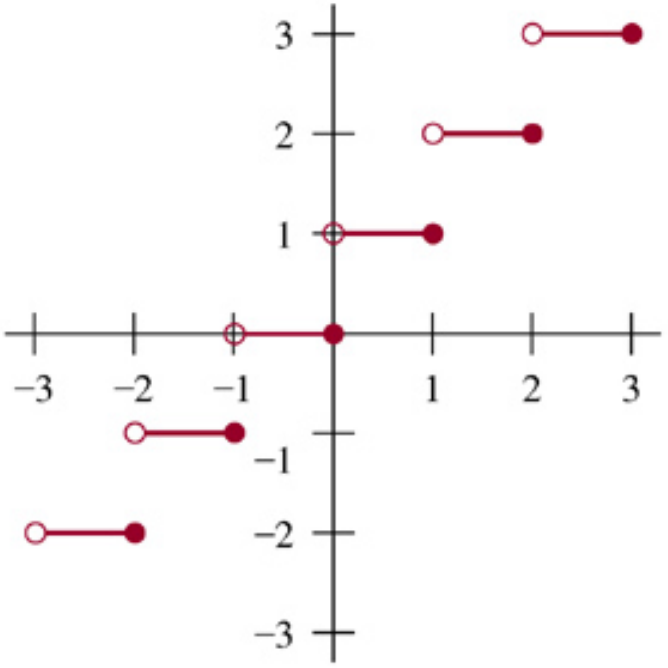
$$\lceil -0.5 \rceil = 0$$



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(a) $y = [x]$



(b) $y = [x]$

Useful Properties of the Floor and Ceiling Functions

(1a) $\lfloor x \rfloor = n$ iff $n \le x < n+1$ where n is an integer

(1b) $\lceil x \rceil = n$ iff $n-1 < x \le n$ where n is an integer

(1c) $\lfloor x \rfloor = n$ iff $x-1 < n \le x$ where n is an integer

(1d) $\lceil x \rceil = n$ iff $x \le n < x+1$ where n is an integer

(2) $x-1 < \lfloor x \rfloor \le x \le \lceil x \rceil < x+1$

(3a) $\lfloor -x \rfloor = -\lceil x \rceil$

(3b) $\lceil -x \rceil = -\lfloor x \rfloor$

(4a) $\lfloor x+m \rfloor = \lfloor x \rfloor + m$ where m is an integer

(4b) $\lceil x+m \rceil = \lceil x \rceil + m$ where m is an integer



Proving Properties of Functions

Example: Prove that x is a real number, then

$$\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + 1/2 \rfloor$$

Solution: Let $x = n + \varepsilon$, where n is an integer and $0 \leq \varepsilon < 1$.

Case 1: $\varepsilon < ½$

$2x = 2n + 2\varepsilon$ and $\lfloor 2x \rfloor = 2n$, since $0 \leq 2\varepsilon < 1$.

$\lfloor x + 1/2 \rfloor = n$, since $x + ½ = n + (1/2 + \varepsilon)$ and $0 \leq ½ + \varepsilon < 1$.

Hence, $\lfloor 2x \rfloor = 2n$ and $\lfloor x \rfloor + \lfloor x + 1/2 \rfloor = n + n = 2n$.

Case 2: $\varepsilon \geq ½$

- $2x = 2n + 2\varepsilon = (2n + 1) + (2\varepsilon - 1)$ and $\lfloor 2x \rfloor = 2n + 1$, since $0 \leq 2\varepsilon - 1 < 1$.
- $\lfloor x + 1/2 \rfloor = \lfloor n + (1/2 + \varepsilon) \rfloor = \lfloor n + 1 + (\varepsilon - 1/2) \rfloor = n + 1$ since $0 \leq \varepsilon - 1/2 < 1$.
- Hence, $\lfloor 2x \rfloor = 2n + 1$ and $\lfloor x \rfloor + \lfloor x + 1/2 \rfloor = n + (n + 1) = 2n + 1$.



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【Definition】 A *sequence* is a function from a subset of the set of integers (usually either the set {0, 1, 2, …} or the set {1, 2, 3, …}) to a set S. We use the notation a_n to denote the image of the integer *n*. We call a_n a term of the sequence.

Note:

- **Notation of sequence: $\{a_n\}$**
- **sequence of $\{a_n\}$: $a_1, a_2, \ldots, a_n, \ldots$**
- **The order in a sequence matters.**



Geometric Progression

A *geometric progression* is a sequence of the form

$$a, ar, ar^2, \ldots, ar^n, \ldots$$

where the initial term *a* and the common ratio *r* are real numbers.

〖Example 1〗 The sequence of $\{a_n\}$ with $a_n = 2 \cdot 5^n$ is a geometric progression with initial term and common ratio equal to 2 and 5 respectively, if we start at $n = 0$. Thus, we can express this sequence with

2, 10, 50, 250, 1250, …



Arithmetic Progression

An *arithmetic progression* is a sequence of the form

$$a, a+d, a+2d, \ldots, a+nd, \ldots$$

where the initial term *a* and the common difference *d* are real numbers.

〖Example 2〗 The sequence of $\{a_n\}$ with $a_n = -1+4n$ is an arithmetic progression with initial term and common difference equal to -1 and 4 respectively, if we start at $n = 0$. Thus, we can express this sequence with

-1, 3, 7, 11, …,



Special Integer Sequences

A common problem in discrete mathematics is finding a formula or a general rule for constructing the terms of a sequence.

〖Example 3〗 How can we produce the terms of a sequence if the first 10 terms are 5, 11, 17, 23, 29, 35, 41, 47, 53, 59?



Special Integer Sequences

- **Deduce a possible formula or rule for the terms of a sequence from the initial terms.**
  - **Are there runs of the same value?**
  - **Are terms obtained from previous terms by adding the same amount or an amount that depends on the position in the sequence?**
  - **Are terms obtained from previous terms by multiplying by a particular amount?**
  - **Are terms obtained by combining previous terms in a certain way?**
  - **Are there cycles among the terms?**



Special Integer Sequences

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TABLE 1 Some Useful Sequences.

nth Term	*First 10 Terms*
n^2	1, 4, 9, 16, 25, 36, 49, 64, 81, 100, . . .
n^3	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, . . .
n^4	1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000, . . .
2^n	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024, . . .
3^n	3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049, . . .
$n!$	1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800, . . .



Summations

$$\sum_{i=m}^{n} a_i = \sum_{i=m}^{n} a_i = \sum_{\mathrm{m}\le\mathrm{i}\le\mathrm{n}} a_j$$

$$= a_m + a_{m+1} + \cdots + a_n$$

Note:

- ***i*: index of summation**
- ***m*: lower limit**
- ***n*: upper limit**



Summations

$$\sum_{s \in S} f(s)$$

S: the subset of the domain of the function f

〖Example 4〗

$$\sum_{s \in \{0,2,4\}} s = 0 + 2 + 4 = 6$$



TABLE 2 Some Useful Summation Formulae.

Sum	*Closed Form*
$\sum_{k=0}^{n} ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^{n} k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^{n} k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^{n} k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, \lvert x \rvert < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty}, kx^{k-1}, \lvert x \rvert < 1$	$\frac{1}{(1-x)^2}$

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The cardinality of an infinite set

Some interesting questions:

1) **How to count the number of elements in an infinite set?**
2) **Are the cardinalities of all infinite sets same?**
3) **Is the number of positive integers double the number of positive even integers?**
4) **Is the number of real numbers in (a,b) more than the number of real numbers in (0,1)?**
5) **Are there same number of rational numbers and real numbers in (0,1)?**



【Definition 1】 The cardinality of a set A is equal to the cardinality of a set B, denoted $|A| = |B|$, iff there exists a bijection from A to B.

Question 1

【Definition 2】 If there is an injection from A to B, the cardinality of A is less than or the same as the cardinality of B and we write $|A| \leq |B|$. When $|A| \leq |B|$ and A and B have different cardinality, we say that the cardinality of A is less than the cardinality of B and write $|A| < |B|$.

〖Example 1〗 let A be the set of 26 lowercase English alphabets. B={1,2,…,26}. Then $|A| = |B|$.

〖Example 2〗 The set of positive integers is denoted by $Z^+ = \{1,2,\cdots,n,\cdots\}$, $M = \{1,2^2,\cdots,n^2,\cdots\}$ Then $|Z^+| = |M|$.



〖Example 3〗 Let A be the set of real numbers between a and b ($a<b$), and B be the set of real numbers between 0 and 1. Show that $|A| = |B|$.

Question 4

Proof:

Let f be a function from A to B.

$$\frac{x-a}{b-a}=\frac{y-0}{1-0}$$

$$y=f(x)=\frac{x-a}{b-a}$$

Then y is a bijection from (a, b) to $(0,1)$.

Hence, $|A| = |B|$.



【Definition】A set that is either finite or has the same cardinality as the set of positive integers called *countable*.

A set that is not countable is called *uncountable*.

When an infinite set S is countable, we denote the cardinality of S by $\aleph_0$ (*aleph null*).

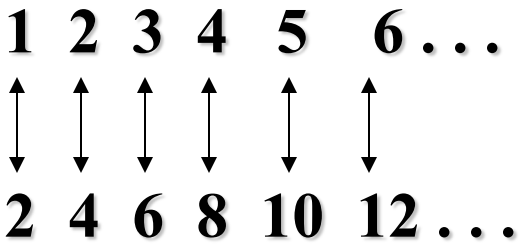
If $|A| = |Z^+|$, the set A is *countable infinite*.



Some Countable Infinite Sets

1) The set E of even positive integers

Let $f(x) = 2x$. Then, f is a bijection from Z^+ to E .



Question 3

Note :

- E is countable infinite.
- The numbers of positive even integers is the same as positive integers
- E is a proper subset of Z^+, but $|E| = |Z^+|$.

2) The set of integers Z

$$Z = \{0,1,-1,2,-2,\cdots\}$$

$$f(n) = \begin{cases} -(n-1)/2 & n \text{ is odd} \\ n/2 & n \text{ is even} \end{cases}$$

***f* is a bijection from Z^+ to Z .**

Hence, Z is countable infinite set。



3) The set of positive rational numbers Q^+

- $\forall x \in Q^+, x = p/q,\ p, q \in Z^+$
- Let $S=\{ (p, q) \mid p, q \in Z^+ \} = Z^+ \times Z^+$.
- $$\left.\begin{array}{l} |Q^+| \le |S| \\ |S| = |Z^+| \\ |Z^+| \le |Q^+| \end{array}\right\} \Rightarrow |Q^+| = |Z^+|$$



(1) $|Q^+| \leq |S|$

Suppose that $r = \frac{p}{q} \in Q^+$

$\frac{p}{q} \rightarrow (p, q)$ is injective

$\therefore |Q^+| \leq |S|$



(2) $|S| = |Z^+|$ **An infinite set is countable iff it is possible to list all the elements of the set in a sequence**

	1	2	3	…	*p*	…
1	(1,1)	(2,1)	(3,1)	…	(*p*,1)	…
2	(1,2)	(2,2)	(3,2)	…	(*p*,2)	…
3	(1,3)	(2,3)	(3,3)	…	(*p*,3)	…
…	…					
q	(1,*q*)	(2,*q*)	(3,*q*)	…	(*p*,*q*)	…
…	…					

$$1+2+\cdots+(p+q-2)=\frac{(p+q-2)(p+q-1)}{2}$$

$$n=\frac{1}{2}(p+q-2)(p+q-1)+q$$



(3) $|Z^+| \leq |Q^+|$

$\because Z^+ \subseteq Q^+$

$\therefore |Z^+| \leq |Q^+|$

Note :

(1) There are the same number of positive rational numbers and positive integers.

(2) The set of all rational numbers Q, positive and negative, is countable infinite.



The properties of the countable sets:

1) **No infinite set has a smaller cardinality than a countable set.**
2) **The union of two countable sets is countable.**
3) **The union of finite number of countable sets is countable.**
4) **The union of a countable number of countable sets is countable.**



3. Cantor Diagonalization Argument

---- **An important technique used to construct an object which is not a member of a countable set of objects**

【Theorem】 The set of real numbers between 0 and 1 is uncountable.

Proof:

$$A = \{x \mid x \in (0,1) \wedge x \in R\}$$

$$(1) \mid Z^+ \mid \leq \mid A \mid$$

$$(2) \mid Z^+ \mid \neq \mid A \mid$$

$$\Rightarrow \quad \mid Z^+ \mid < \mid A \mid$$

Question 2



(1) $|Z^+| \leq |A|$

$$A = \{x \mid x \in (0,1) \wedge x \in R\}$$

$$B = \{\frac{1}{n+1} \mid n \in Z^+\}$$

$\because |B| = |Z^+|$ $\quad\quad$ $B \subseteq A$

$\therefore |Z^+| \leq |A|$



(2) $|Z^+| \neq |A|$

Assume A is countable, then let A={r_1 , r_2 , r_3 ,… , r_n ,… }

Represent each real number in the list using *its decimal expansion*.

e.g.,1/3 = .3333333....., 1/2 = .5000000.....= .4999999.....

THE LIST....

$r_1 = 0.d_{11}d_{12}d_{13}d_{14}d_{15}d_{16}\ldots.$

$r_2 = 0.d_{21}d_{22}d_{23}d_{24}d_{25}d_{26}\ldots.$

$r_3 = 0.d_{31}d_{32}d_{33}d_{34}d_{35}d_{36}\ldots.$

…

Now construct the number $x = 0.x_1x_2x_3x_4x_5x_6x_7\ldots.$

$x_i = 3$ if $d_{ii} \neq 3$

$x_i = 4$ if $d_{ii} = 3$

Then x is not equal to any number in the list.

Hence, no such list can exist and hence the interval (0,1) is uncountable .



【Theorem】The set of real numbers $R=(-\infty,+\infty)$ has the same cardinality as the set (0,1).

Proof:

Let *f*(*x*)=tg(*x*).

$f(x)$ is a bijection from $(-\frac{\pi}{2},\frac{\pi}{2})$ to $R=(-\infty,+\infty)$.

$$\because \left|(-\frac{\pi}{2},\frac{\pi}{2})\right| = |(0,1)| \qquad \therefore |R| = |(0,1)|$$

$|R|=\aleph_1$



〖Example 4〗 Suppose that $[0,1]=\{x \mid x \in R, 0 \le x \le 1\}$ **. Show that the cardinality of this set is** $\aleph_1$**.**

Proof:

$$A=[0,1]=\{x \mid x \in R, 0 \le x \le 1\}$$

$$B=(0,1)=\{x \mid x \in R, 0 < x < 1\}$$

$$(1) B \subseteq A \Rightarrow |B| \le |A|$$

$$(2) \text{ Let } g(x)=\frac{1}{2}x+\frac{1}{4}, x \in [0,1]$$

Hence, *g*(*x*) is a bijection from [0,1] to [1/4,3/4].

$$\text{Thus } |A| \le |B|$$



〖Example 5〗 Show that the set of finite strings S over a finite alphabet A is countably infinite.

Proof:

Assume an alphabetical ordering of symbols in A. Show that the strings can be listed in a sequence. First list

1. **All the strings of length 1 in alphabetical order.**
2. **Then all the strings of length 2 in lexicographic (as in a dictionary) order.**
3. **Then all the strings of length 3 in lexicographic order.**
4. **And so on.**

This implies a bijection from N to S and hence it is a countably infinite set.



〖Example 6〗 Show that the set of all Java programs is countable.



Computability

We say that a function is computable if there is a computer program in some programming language that finds the values of this function. If a function is not computable we say it is uncomputable.

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【Schrőder-Bernstein Theorem】If A and B are sets with $|A| \leq |B|$ and $|B| \leq |A|$ then $|A| = |B|$. In other words, if there are one-to-one functions f from A to B and g from B to A, then there is a one to –one correspondence between a A and B.

【Theorem】The cardinality of the power set of an arbitrary set has a greater cardinality than the original arbitrary set.

The Continuum Hypothesis

The continuum hypothesis (CH) asserts that there is no cardinal number a such that $\aleph_0 < a < \aleph_1$.



Homework: (Due on March 28)

Ver. 7

Sec. 2.3 22(c), 34, 40(a), 72, 74(c,d)

Sec. 2.5 4(c,d), 28 , 36, 38

Ver. 8

Sec. 2.3 22(c), 36, 42(a), 74, 76(c,d)

Sec. 2.5 4(c,d), 28 , 36, 38

